

General Solution Approach for Convolution Problems

1. Understand the Convolution Operation:

- Convolution involves sliding a filter (also called a kernel) across the input data (which could be a vector or a matrix).
- At each position, you perform element-wise multiplication between the filter and the input data's corresponding section, then sum up the results to produce a single output value.

2. Determine the Size of the Output:

- 1D Convolution: For a 1D input vector and a 1D filter, the output size is determined using the formula:

$$\{\text{Output size}\} = \{\text{Input size}\} - \{\text{Filter size}\} + 1$$

- 2D Convolution: For a 2D input matrix and a 2D filter, the output size is calculated similarly and applied in both dimensions:

$$\{\text{Output width/height}\} = \{\text{Input width/height}\} - \{\text{Filter width/height}\} + 1$$

- Ensure you account for any stride (the number of pixels/units the filter moves at each step) and padding (extra borders added to the input).

3. Apply the Filter:

- 1D Convolution Example: For a filter of size 1, simply multiply each element of the input vector by the filter value.
- 2D Convolution Example: Place the filter on top of the input matrix and slide it across it. At each position, perform the element-wise multiplication, sum the results, and

place the sum in the corresponding position in the output matrix.

4. Interpret the Results

The convolution output represents a transformation of the input data based on the filter applied. For example, certain filters can detect edges in an image or amplify specific patterns in the data.

5. Check for Special Cases

- If the stride is greater than 1, the output size will be smaller as the filter moves more than one unit at a time.
- If padding is added, the output size may be larger or equal to the input size, depending on the amount of padding.

Applying the General Approach

Problem 1 (1D Convolution)

Multiply each element of the input vector by the filter value. The output vector will be the same size as the input vector since the filter size is 1.

Problem 2 (Size of Output after Convolution)

Use the formula for output size and apply it to the input and filter dimensions provided.

Problem 3 (2D Convolution with Stride)

Calculate the output size by accounting for the stride and apply the convolution process accordingly.